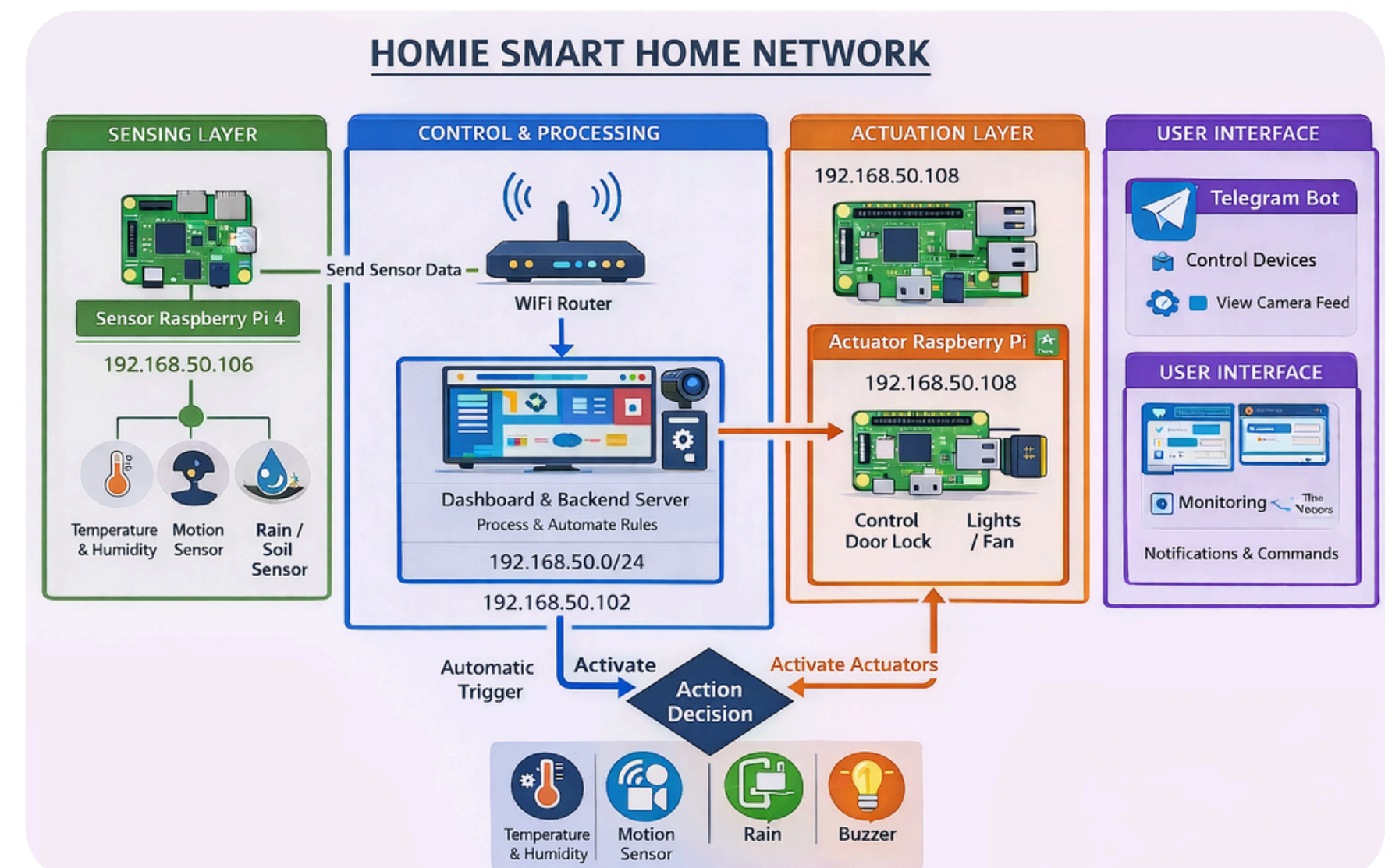


HOMIE: A Network-Integrated Smart Home IoT Monitoring and Control System

Project Overview

HOMIE is a smart home system that uses networking to connect multiple devices for real-time monitoring and control.

- Sensor Data (temperature, motion, rain) is sent via REST APIs to a central dashboard, where it is displayed and used to control actuators such as LEDs and Servos
- The system also integrates external weather APIs to provide smart alerts, demonstrating both local and internet-based communication using TCP/IP



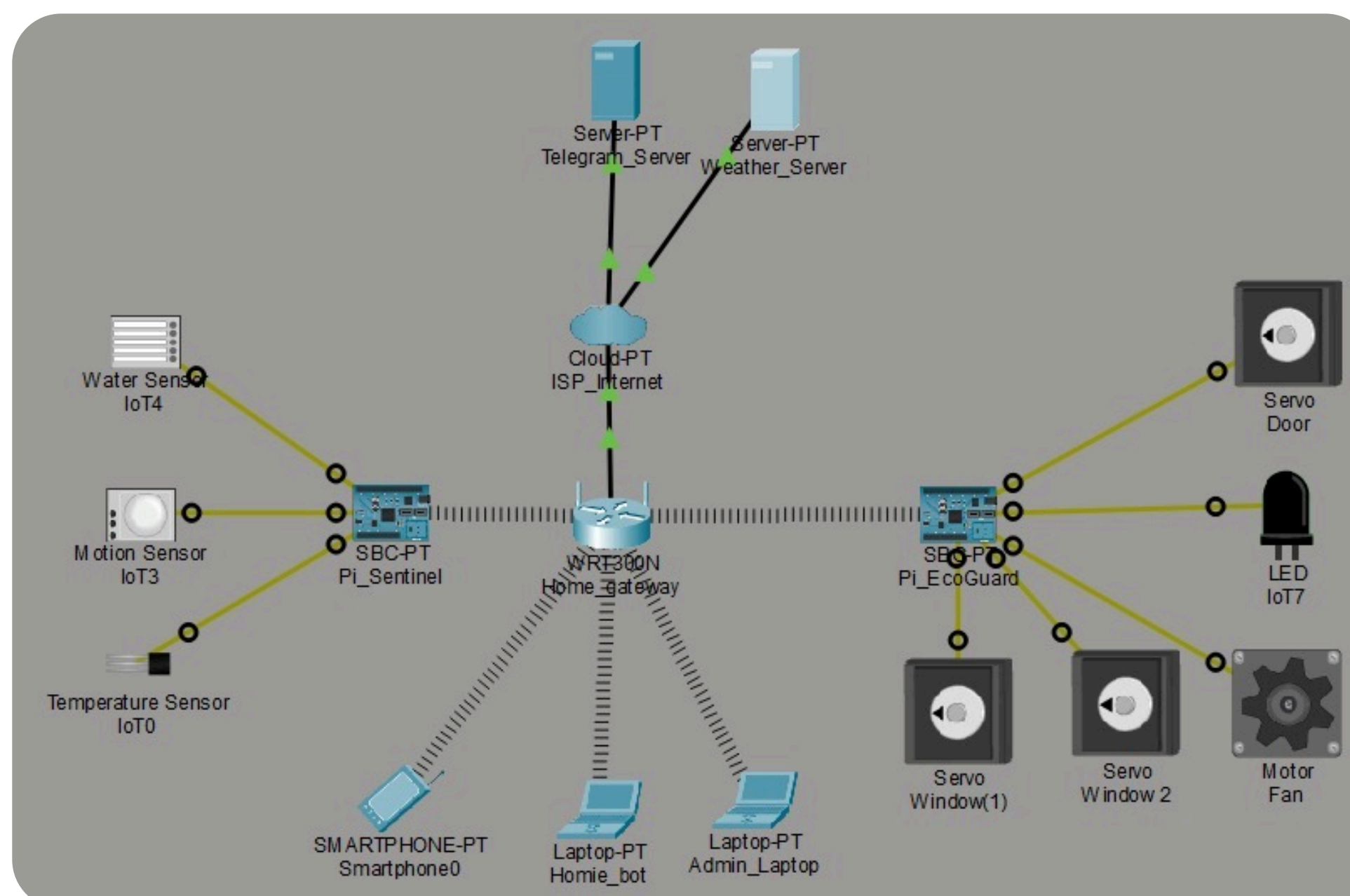
Network Diagram

Hardware

- Raspberry Pi
- PIR sensor
- 4x Servo Motor
- LED
- Soil moisture

Software

- JavaScript
- Flask
- Python



Rain Sensor

- Detect rain conditions
- Window/Servo closes when conditions met

Temperature Sensor

- Detect ambient temperature
- Fan on and windows open

PIR Sensor

- Detect movement
- LED On

Camera

- Scan face for known user
- Unlock door if recognised, alert if unknown

System Overview

Real-Time Monitoring

- live sensor data via LAN
- Dashboard updates accordingly

Automated Controls

- Sensor-triggered actions
- Multi-threshold logic

Network Communication

- REST APIs over TCP/IP
- Reliable data exchange across devices

Smart Decision-making and external integration

- Combines multiple sensor inputs
- Executes action automatically
- Weather API support
- Smart alerts

